MANAGEMENT OPERATING REVIEW - RECORD 012

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Quarterly Business Review for Q4 2024: purpose and attendance

Data definitions travel with the packet now, because two functions were counting the same thing differently. Owen Castellanos, VP Customer Success, presented the operations section. Notes were circulated on 2025-02-05. Annual planning was folded into this session. Next year's plan was already signed. Pre-reads went out ahead of the 2025-02-05 session. One function submitted its pre-read late. Headcount 183 of 194, Owen Castellanos reporting. Yield 92 percent, scrap 25 thousand in Harness and electrical. On-time delivery 92 percent, Brightwater Motion the outlier. Crew Foxtrot closed Tulsa without a second visit.

Attendance was complete across the functions that own the numbers, and two guests joined for the sections that concerned them. Nadia Oyelaran attended as Chief of Staff. The next session is set for 2025-02-27. Annual planning was folded into this session. Next year's plan was already signed. The Quarterly Business Review convened for Q4 2024. One function submitted its pre-read late. Headcount 183 of 194, Owen Castellanos reporting. Burn 8910 thousand, runway 18 months. Days outstanding 51, Tenmile Freight the largest balance. Utilisation 91 percent, Crew Foxtrot fully committed.

Meeting discipline held: sections that had no decision to make were noted as read and skipped in the room. Pre-reads went out ahead of the 2025-02-05 session. Owen Castellanos, VP Customer Success, presented the operations section. Annual planning was folded into this session. Next year's plan was already signed. One function submitted its pre-read late. The next session is set for 2025-02-27. Slipped deals 14, including Marrowfield Grocers. Toronto depot cleared its repair backlog. Tessellate Controls quoted lead times moved again. Priya Ramanathan and Dana Whitlock disagreed on sequencing.

Notes were captured live and circulated the same day, which ended a habit of arguing about what was decided. The next session is set for 2025-02-27. Owen Castellanos, VP Customer Success, presented the operations section. Annual planning was folded into this session. Next year's plan was already signed. The Quarterly Business Review convened for Q4 2024. Nadia Oyelaran attended as Chief of Staff. Uptime 97.3 percent, interventions every 344 hours. Toronto depot cleared its repair backlog. Bookings 204 units across 118 pursuits. Elk Grove carries 20 of the queued fleet.

Order book and bookings quality

Regional coverage remains uneven, with one seller carrying a territory that the coverage model says needs two. Intake closed at 204 units for Q4 2024. Win rate held at 34 percent on 118 live opportunities. Annual planning was folded into this session. Next year's plan was already signed. Open pursuits number 118, up from 105. Signed volume: 204 units, win rate 34 percent. Headcount 183 of 194, Owen Castellanos reporting. Yield 92 percent, scrap 25 thousand in Harness and electrical. On-time delivery 92 percent, Brightwater Motion the outlier. Crew Foxtrot closed Tulsa without a second visit.

Nobody in commercial believes the current forecast discipline is good enough, and the review agreed to inspect deal-level notes rather than roll-ups. Open pursuits number 118, up from 105. Discount depth peaked on the Marrowfield Grocers multi-site award. Annual planning was folded into this session. Next year's plan was already signed. Intake closed at 204 units for Q4 2024. Win rate held at 34 percent on 118 live opportunities. Headcount 183 of 194, Owen Castellanos reporting. Burn 8910 thousand, runway 18 months. Days outstanding 51, Tenmile Freight the largest balance. Utilisation 91 percent, Crew Foxtrot fully committed.

The competitive picture shifted, with two losses attributed to a rival that had not appeared in a deal review for three periods. Tenmile Freight accounted for the largest single order. Intake closed at 204 units for Q4 2024. Annual planning was folded into this session. Next year's plan was already signed. The book took 204 units, against 58 last period. Discount depth peaked on the Marrowfield Grocers multi-site award. Slipped deals 14, including Marrowfield Grocers. Toronto depot cleared its repair backlog. Tessellate Controls quoted lead times moved again. Priya Ramanathan and Dana Whitlock disagreed on sequencing.

Commercial intake for the period came in ahead of the internal plan on unit count and behind it on gross margin, which is the trade the team was asked to make. Tenmile Freight accounted for the largest single order. The book took 204

units, against 58 last period. Annual planning was folded into this session. Next year's plan was already signed. Win rate held at 34 percent on 118 live opportunities. Open pursuits number 118, up from 105. Uptime 97.3 percent, interventions every 344 hours. Toronto depot cleared its repair backlog. Bookings 204 units across 118 pursuits. Elk Grove carries 20 of the queued fleet.

Discounting concentrated in one competitive displacement rather than spreading across the book, which the review treated as an acceptable outcome. Tenmile Freight accounted for the largest single order. Intake closed at 204 units for Q4 2024. Annual planning was folded into this session. Next year's plan was already signed. Discount depth peaked on the Marrowfield Grocers multi-site award. Win rate held at 34 percent on 118 live opportunities. Scrap 25 thousand, Penrose Castings contributing. Build plan 72, actual 70, Controllers short. Yield 92 percent, scrap 25 thousand in Harness and electrical. Work in process 30 units, mostly Controllers.

Late-quarter intake distorted the shape of the funnel again, and the pattern is now visible in three consecutive periods of data. Tenmile Freight accounted for the largest single order. Open pursuits number 118, up from 105. Annual planning was folded into this session. Next year's plan was already signed. The book took 204 units, against 58 last period. Signed volume: 204 units, win rate 34 percent. Requisitions open 25, Nadia Oyelaran holding the list. Toronto depot cleared its repair backlog. Go-live cycle 54 days at Redlands. Sunil Achterberg owns the Controllers recovery.

Headcount plan against actual by function and location, Q4 2024

Function	Location	Plan	Actual	Gap	Open requisitions
Engineering	headquarters and plant	11	5	10	6
Manufacturing	headquarters and plant	5	34	0	4
Deployment and field service	headquarters and plant	4	26	0	6
Commercial	headquarters and plant	1	20	0	3
Customer success	headquarters and plant	37	15	35	2
Supply chain	headquarters and plant	26	7	30	2
Finance and administration	headquarters and plant	21	5	25	1
People	headquarters and plant	14	4	16	1
Engineering	field and regional	6	2	5	5
Manufacturing	field and regional	3	18	0	3
Deployment and field service	field and regional	2	13	0	5
Commercial	field and regional	1	10	0	2
Customer success	field and regional	20	7	17	1
Supply chain	field and regional	15	4	15	1
Finance and administration	field and regional	12	3	12	0
People	field and regional	8	2	8	0

Pursuit review

Of the opportunities that left the quarter, most remain live and dated, and only a minority were lost to a competitor outright. Conversion into commercial close remains the weak stage. Owen Castellanos owns the re-qualification pass. Annual planning was folded into this session. Next year's plan was already signed. Deals leaving the period: 14, most still dated. Weighted coverage sits near 118 open pursuits. Headcount 183 of 194, Owen Castellanos reporting. Yield 92 percent, scrap 25 thousand in Harness and electrical. On-time delivery 92 percent, Brightwater Motion the outlier. Crew Foxtrot closed Tulsa without a second visit.

Two accounts that were treated as sure things at the last review have gone quiet, and neither seller has escalated. Slippage totalled 14 opportunities against 7 before. The Havering Logistics pursuit re-dated again. Annual planning was folded into this session. Next year's plan was already signed. Conversion into commercial close remains the weak stage. Deals leaving the period: 14, most still dated. Headcount 183 of 194, Owen Castellanos reporting. Burn 8910 thousand, runway 18 months. Days outstanding 51, Tenmile Freight the largest balance. Utilisation 91 percent, Crew Foxtrot fully committed.

Where a pilot converted, the common thread was an operations sponsor who owned a throughput target rather than a technology budget. The Havering Logistics pursuit re-dated again. Slippage totalled 14 opportunities against 7 before. Annual planning was folded into this session. Next year's plan was already signed. Deals leaving the period: 14, most still dated. Weighted coverage sits near 118 open pursuits. Slipped deals 14, including Marrowfield Grocers. Toronto depot cleared its repair backlog. Tessellate Controls quoted lead times moved again. Priya Ramanathan and Dana Whitlock disagreed on sequencing.

Forecast accuracy at the regional level was worse than the aggregate, meaning two regions cancelled each other's errors out. Conversion into commercial close remains the weak stage. Two losses went to the same competitor. Annual planning was folded into this session. Next year's plan was already signed. Owen Castellanos owns the re-qualification pass. Slippage totalled 14 opportunities against 7 before. Uptime 97.3 percent, interventions every 344 hours. Toronto depot cleared its repair backlog. Bookings 204 units across 118 pursuits. Elk Grove carries 20 of the queued fleet.

Stage conversion tells a cleaner story than the headline number does, because the drop is concentrated between technical validation and commercial close. Deals leaving the period: 14, most still dated. Two losses went to the same competitor. Annual planning was folded into this session. Next year's plan was already signed. Conversion into commercial close remains the weak stage. Weighted coverage sits near 118 open pursuits. Scrap 25 thousand, Penrose Castings contributing. Build plan 72, actual 70, Controllers short. Yield 92 percent, scrap 25 thousand in Harness and electrical. Work in process 30 units, mostly Controllers.

Deal inspection surfaced a habit of re-dating rather than re-qualifying, and the review asked for a hard rule on how many times a close date can move. Owen Castellanos owns the re-qualification pass. Two losses went to the same competitor. Annual planning was folded into this session. Next year's plan was already signed. Deals leaving the period: 14, most still dated. The Havering Logistics pursuit re-dated again. Requisitions open 25, Nadia Oyelaran holding the list. Toronto depot cleared its repair backlog. Go-live cycle 54 days at Redlands. Sunil Achterberg owns the Controllers recovery.

Requisition ageing and decisions, Q4 2024

Requisition	Role	Days open	Stage	Recruiting owner	Decision this review
REQ-1240	Field engineer	70	on hold	hiring manager sourced	continue
REQ-1242	Crew lead	99	closed without hire	external agency	raise band
REQ-1244	Controls engineer	56	sourcing	internal recruiting	widen radius
REQ-1246	Manufacturing technician	89	phone screen	external agency	pause
REQ-1248	Buyer and planner	70	on-site loop	internal recruiting	close and re-scope
REQ-1250	Solutions architect	99	offer out	external agency	convert contractor
REQ-1252	Service dispatcher	58	offer declined	external agency	continue
REQ-1254	Reliability engineer	79	on hold	internal recruiting	raise band
REQ-1256	Field engineer (second opening)	52	closed without hire	Renata Voss	widen radius
REQ-1258	Crew lead (second opening)	85	sourcing	hiring manager sourced	pause
REQ-1260	Controls engineer (second opening)	66	phone screen	external agency	close and re-scope
REQ-1262	Manufacturing technician (second opening)	95	on-site loop	hiring manager sourced	convert contractor
REQ-1264	Buyer and planner (second opening)	70	offer out	external agency	continue
REQ-1266	Solutions architect (second opening)	99	offer declined	external agency	raise band
REQ-1268	Service dispatcher (second opening)	56	on hold	external agency	widen radius
REQ-1270	Reliability engineer (second opening)	89	closed without hire	hiring manager sourced	pause
REQ-1272	Field engineer (backfill)	70	sourcing	external agency	close and re-scope

Installation programme

Sequencing the queue by contractual date rather than by readiness produced two expensive standstills, and the rule has been reversed. Nadia Oyelaran sequenced the queue by readiness. Chandler South and Redlands both cleared

acceptance testing. Annual planning was folded into this session. Next year's plan was already signed. Queue depth reached 20 sites. Go-live cycle: 54 days, from 61. Headcount 183 of 194, Owen Castellanos reporting. Yield 92 percent, scrap 25 thousand in Harness and electrical. On-time delivery 92 percent, Brightwater Motion the outlier. Crew Foxtrot closed Tulsa without a second visit.

A retrofit site behaved nothing like a greenfield site, and the estimate model still treats them as the same class of work. Go-live cycle: 54 days, from 61. Queue depth reached 20 sites. Annual planning was folded into this session. Next year's plan was already signed. Elk Grove was held at the readiness gate. The queue carries 20 sites, 21 a period ago. Headcount 183 of 194, Owen Castellanos reporting. Burn 8910 thousand, runway 18 months. Days outstanding 51, Tenmile Freight the largest balance. Utilisation 91 percent, Crew Foxtrot fully committed.

Post-go-live stabilisation periods are running longer than the plan allows, so crews are being pulled back to sites already signed off. Chandler South and Redlands both cleared acceptance testing. Elk Grove was held at the readiness gate. Annual planning was folded into this session. Next year's plan was already signed. Queue depth reached 20 sites. The queue carries 20 sites, 21 a period ago. Slipped deals 14, including Marrowfield Grocers. Toronto depot cleared its repair backlog. Tessellate Controls quoted lead times moved again. Priya Ramanathan and Dana Whitlock disagreed on sequencing.

Site queue depth grew again, and the queue is now long enough that sequencing decisions have commercial consequences rather than just scheduling ones. Nadia Oyelaran sequenced the queue by readiness. Chandler South and Redlands both cleared acceptance testing. Annual planning was folded into this session. Next year's plan was already signed. Go-live cycle: 54 days, from 61. Contract to go-live averaged 54 days. Uptime 97.3 percent, interventions every 344 hours. Toronto depot cleared its repair backlog. Bookings 204 units across 118 pursuits. Elk Grove carries 20 of the queued fleet.

Readiness surveys are now a hard gate, and two sites were held at the gate rather than allowed to start with known network gaps. Go-live cycle: 54 days, from 61. Contract to go-live averaged 54 days. Annual planning was folded into this session. Next year's plan was already signed. The queue carries 20 sites, 21 a period ago. Nadia Oyelaran sequenced the queue by readiness. Scrap 25 thousand, Penrose Castings contributing. Build plan 72, actual 70, Controllers short. Yield 92 percent, scrap 25 thousand in Harness and electrical. Work in process 30 units, mostly Controllers.

Customer-side readiness remains the single largest determinant of go-live date, and it is the variable the company controls least. Queue depth reached 20 sites. Nadia Oyelaran sequenced the queue by readiness. Annual planning was folded into this session. Next year's plan was already signed. Contract to go-live averaged 54 days. Go-live cycle: 54 days, from 61. Requisitions open 25, Nadia Oyelaran holding the list. Toronto depot cleared its repair backlog. Go-live cycle 54 days at Redlands. Sunil Achterberg owns the Controllers recovery.

Deployment resourcing

Contractor quality varied by firm rather than by individual, and the sourcing decision now reflects that. Installer utilisation ran at 91 percent. Sunil Achterberg covered two regions this period. Annual planning was folded into this session. Next year's plan was already signed. Certified crew count trails the site queue of 20. Tulsa needed a second crew visit. Headcount 183 of 194, Owen Castellanos reporting. Yield 92 percent, scrap 25 thousand in Harness and electrical. On-time delivery 92 percent, Brightwater Motion the outlier. Crew Foxtrot closed Tulsa without a second visit.

The apprenticeship route produced its first fully certified engineer, which took longer than planned and cost less than lateral hiring. Crew leads remain the gating resource. Utilisation was 91 percent against 94 percent. Annual planning was folded into this session. Next year's plan was already signed. Installer utilisation ran at 91 percent. Tulsa needed a second crew visit. Headcount 183 of 194, Owen Castellanos reporting. Burn 8910 thousand, runway 18 months. Days outstanding 51, Tenmile Freight the largest balance. Utilisation 91 percent, Crew Foxtrot fully committed.

Deployment capacity is a function of certified crew leads, not of headcount, and the certification pipeline is the part nobody was watching. Sunil Achterberg covered two regions this period. Utilisation was 91 percent against 94 percent. Annual planning was folded into this session. Next year's plan was already signed. Tulsa needed a second crew visit. Travel days per site fell again. Slipped deals 14, including Marrowfield Grocers. Toronto depot cleared its repair backlog. Tessellate Controls quoted lead times moved again. Priya Ramanathan and Dana Whitlock disagreed on sequencing.

Travel load on the crew has become a retention issue, and the two most recent resignations both named it in an exit conversation. Sunil Achterberg covered two regions this period. Installer utilisation ran at 91 percent. Annual planning was folded into this session. Next year's plan was already signed. Crew leads remain the gating resource. Utilisation was 91 percent against 94 percent. Uptime 97.3 percent, interventions every 344 hours. Toronto depot cleared its repair backlog. Bookings 204 units across 118 pursuits. Elk Grove carries 20 of the queued fleet.

Certification standards were tightened rather than relaxed under schedule pressure, and the review recorded that choice deliberately. Utilisation was 91 percent against 94 percent. Installer utilisation ran at 91 percent. Annual planning was folded into this session. Next year's plan was already signed. Certified crew count trails the site queue of 20. Sunil Achterberg covered two regions this period. Scrap 25 thousand, Penrose Castings contributing. Build plan 72, actual 70, Controllers short. Yield 92 percent, scrap 25 thousand in Harness and electrical. Work in process 30 units, mostly Controllers.

Utilisation above ninety percent leaves nothing for the unplanned second visit, and the second visit happens on roughly one site in four. Installer utilisation ran at 91 percent. Tulsa needed a second crew visit. Annual planning was folded into this session. Next year's plan was already signed. Travel days per site fell again. Sunil Achterberg covered two regions this period. Requisitions open 25, Nadia Oyelaran holding the list. Toronto depot cleared its repair backlog. Go-live cycle 54 days at Redlands. Sunil Achterberg owns the Controllers recovery.

Leavers and backfill position, Q4 2024

Function	Leavers in period	Annualised attrition %	Primary stated reason	Backfill position
Engineering	0	8	retirement	deferred to next quarter
Manufacturing	2	15	performance	requisition open
Deployment and field service	1	10	travel load	backfilled internally
Commercial	3	17	compensation	not backfilling
Customer success	1	6	relocation	contractor covering
Supply chain	4	14	career change outside the industry	deferred to next quarter
Finance and administration	0	9	manager relationship	requisition open
People	3	16	retirement	backfilled internally
Engineering (contract)	1	7	performance	not backfilling
Manufacturing (contract)	3	12	travel load	contractor covering
Deployment and field service (contract)	0	5	compensation	deferred to next quarter
Commercial (contract)	2	13	relocation	requisition open
Customer success (contract)	0	8	career change outside the industry	backfilled internally

Assembly output and attainment

Build attainment against plan was short again, and the shortfall traces to component availability rather than to line throughput. Plan was 72 units; the line delivered 70. Output reached 70 units, from 60. Annual planning was folded into this session. Next year's plan was already signed. Work in process ended at 30 units. The HS-400 pallet mover drove most of the changeover time. Headcount 183 of 194, Owen Castellanos reporting. Yield 92 percent, scrap 25 thousand in Harness and electrical. On-time delivery 92 percent, Brightwater Motion the outlier. Crew Foxtrot closed Tulsa without a second visit.

Overtime carried the last two weeks of the period, which closed the gap on units and did nothing good for cost per unit. Priya Ramanathan owns the line rebalance. Work in process ended at 30 units. Annual planning was folded into this session. Next year's plan was already signed. The HS-400 pallet mover drove most of the changeover time. Plan was 72 units; the line delivered 70. Headcount 183 of 194, Owen Castellanos reporting. Burn 8910 thousand, runway 18 months. Days outstanding 51, Tenmile Freight the largest balance. Utilisation 91 percent, Crew Foxtrot fully committed.

Manufacturing engineering closed out the top three line stoppages, and the fourth turns out to be a fixture that nobody owns. Overtime carried the closing two weeks. Attainment gap: 2 units short of plan. Annual planning was folded into this session. Next year's plan was already signed. Output reached 70 units, from 60. Priya Ramanathan owns the line rebalance. Slipped deals 14, including Marrowfield Grocers. Toronto depot cleared its repair backlog. Tessellate Controls quoted lead times moved again. Priya Ramanathan and Dana Whitlock disagreed on sequencing.

Plan stability is now the loudest complaint from the plant, with three revisions to the same month's plan in one period. The HS-400 pallet mover drove most of the changeover time. Work in process ended at 30 units. Annual planning was folded into this session. Next year's plan was already signed. Overtime carried the closing two weeks. Priya Ramanathan owns the line rebalance. Uptime 97.3 percent, interventions every 344 hours. Toronto depot cleared its repair backlog. Bookings 204 units across 118 pursuits. Elk Grove carries 20 of the queued fleet.

The plant met its unit number and missed its labour-hours-per-unit target, which the review treated as the more important miss. Attainment gap: 2 units short of plan. Output reached 70 units, from 60. Annual planning was folded into this session. Next year's plan was already signed. Work in process ended at 30 units. Plan was 72 units; the line delivered 70. Scrap 25 thousand, Penrose Castings contributing. Build plan 72, actual 70, Controllers short. Yield 92 percent, scrap 25 thousand in Harness and electrical. Work in process 30 units, mostly Controllers.

Line output has been steady for three periods while the plan has grown, which is a planning problem dressed up as an execution problem. Attainment gap: 2 units short of plan. Overtime carried the closing two weeks. Annual planning was folded into this session. Next year's plan was already signed. Work in process ended at 30 units. Plan was 72 units; the line delivered 70. Requisitions open 25, Nadia Oyelaran holding the list. Toronto depot cleared its repair backlog. Go-live cycle 54 days at Redlands. Sunil Achterberg owns the Controllers recovery.

Carried constraints

Constraint under review: drive-unit lead time, at the dual sourcing decision stage as of this session. Management committed to qualifying Brightwater Motion as a second source for the drive unit, accepting a higher unit cost during overlap in exchange for removing a single point of failure. Toronto depot cleared its repair backlog. Slipped deals 14, including Marrowfield Grocers.

Constraint under review: sensor assembly yield, at the recovery curve stage as of this session. First-pass yield recovered to ninety-two percent and the review agreed to hold hundred percent inspection until three consecutive weeks clear ninety-five. Uptime 97.3 percent, interventions every 344 hours. Queue depth 20 sites, Chandler South at the front.

Buyers are chasing acknowledgements on a fixed cadence now, and the cadence has already caught two silent date pushes. Supplier on-time delivery came in at 92 percent. Expedite activity rose with the build plan of 72. Annual planning was folded into this session. Next year's plan was already signed. Penrose Castings improved after a site visit. On-time delivery was 92 percent, from 89 percent. Harness and electrical ran 92 percent first pass. Tessellate Controls quoted lead times moved again. Yield 92 percent, scrap 25 thousand in Harness and electrical. Elk Grove carries 20 of the queued fleet.

Burn by category, Q4 2024

Category	Period USD thousands	Prior period USD thousands	Movement %	Commentary
Engineering payroll	720	94	666.0	qualification testing
Manufacturing payroll and overheads	623	59	955.9	spares buy for the drive unit
Deployment labour, employees	356	119	199.2	event spend deferred
Deployment labour, contract crews	534	59	805.1	headcount ramp
Commercial payroll	445	89	400.0	contract crews on two sites
Marketing and events	445	119	273.9	spares buy for the drive unit
Field service payroll	356	595	-40.2	contract crews on two sites
Spares and consumables	267	416	-35.8	event spend deferred
Facilities and rent	178	297	-40.1	spares buy for the drive unit
Information technology and tooling	356	208	71.2	qualification testing
Professional services	178	208	-14.4	test cell deferred
Recruiting fees	267	148	80.4	headcount ramp
Travel and subsistence	356	178	100.0	test cell deferred
Insurance and compliance	1692	119	1321.8	headcount ramp
Test and validation materials	1336	148	802.7	contract crews on two sites
Capital expenditure	801	119	573.1	event spend deferred

Quality performance

Root cause work took longer than the containment did, which is the correct order and still frustrating to watch. Dana Whitlock signed the corrective action. First-pass yield finished at 92 percent. Annual planning was folded into this session. Next year's plan was already signed. Scrap totalled 25 thousand dollars. Rework hours concentrated in the sensor cell. Headcount 183 of 194, Owen Castellanos reporting. Yield 92 percent, scrap 25 thousand in Harness and electrical. On-time delivery 92 percent, Brightwater Motion the outlier. Crew Foxtrot closed Tulsa without a second visit.

Rework hours in the sensor cell exceeded the assembly hours for two weeks, which is the clearest signal the process was out of control. Brightwater Motion changed a process without notice. Scrap ran 25 thousand, from 27 thousand. Annual planning was folded into this session. Next year's plan was already signed. Rework hours concentrated in the sensor cell. First-pass yield finished at 92 percent. Headcount 183 of 194, Owen Castellanos reporting. Burn 8910 thousand, runway 18 months. Days outstanding 51, Tenmile Freight the largest balance. Utilisation 91 percent, Crew Foxtrot fully committed.

Defect Pareto shifted after the bond-line fix, and the new top category is a connector seating issue with a simple tooling answer. Dana Whitlock signed the corrective action. Yield was 92 percent against 92 percent. Annual planning was folded into this session. Next year's plan was already signed. Scrap totalled 25 thousand dollars. First-pass yield finished at 92 percent. Slipped deals 14, including Marrowfield Grocers. Toronto depot cleared its repair backlog. Tessellate Controls quoted lead times moved again. Priya Ramanathan and Dana Whitlock disagreed on sequencing.

Nobody wanted to hold shipments, and the review held them anyway on eleven units that had incomplete bond-line records. Rework hours concentrated in the sensor cell. Scrap totalled 25 thousand dollars. Annual planning was folded into this session. Next year's plan was already signed. Yield was 92 percent against 92 percent. First-pass yield finished at 92 percent. Uptime 97.3 percent, interventions every 344 hours. Toronto depot cleared its repair backlog. Bookings 204 units across 118 pursuits. Elk Grove carries 20 of the queued fleet.

Scrap dollars tell a different story from scrap units, because the units being scrapped are the expensive ones. First-pass yield finished at 92 percent. Brightwater Motion changed a process without notice. Annual planning was folded into this session. Next year's plan was already signed. Yield was 92 percent against 92 percent. Rework hours concentrated in the sensor cell. Scrap 25 thousand, Penrose Castings contributing. Build plan 72, actual 70, Controllers short. Yield 92 percent, scrap 25 thousand in Harness and electrical. Work in process 30 units, mostly Controllers.

The corrective action closed with evidence rather than with an assertion, and the evidence was three weeks of clean sampling data. First-pass yield finished at 92 percent. Brightwater Motion changed a process without notice. Annual planning was folded into this session. Next year's plan was already signed. Dana Whitlock signed the corrective action. Scrap ran 25 thousand, from 27 thousand. Requisitions open 25, Nadia Oyelaran holding the list. Toronto depot cleared its repair backlog. Go-live cycle 54 days at Redlands. Sunil Achterberg owns the Controllers recovery.

Supply base performance

A battery cell allocation notice arrived with thirty days of warning, and the planning team had inventory to absorb it. Owen Castellanos issued the first supplier scorecards. Tessellate Controls missed two consecutive commit dates. Annual planning was folded into this session. Next year's plan was already signed. Expedite activity rose with the build plan of 72. Coverage weeks on long-lead parts were extended. Headcount 183 of 194, Owen Castellanos reporting. Yield 92 percent, scrap 25 thousand in Harness and electrical. On-time delivery 92 percent, Brightwater Motion the outlier. Crew Foxtrot closed Tulsa without a second visit.

Inbound freight mode shifted toward air on two part numbers, which protected the build plan and shows up in cost per unit. Owen Castellanos issued the first supplier scorecards. Expedite activity rose with the build plan of 72. Annual planning was folded into this session. Next year's plan was already signed. Tessellate Controls missed two consecutive commit dates. On-time delivery was 92 percent, from 89 percent. Headcount 183 of 194, Owen Castellanos reporting. Burn 8910 thousand, runway 18 months. Days outstanding 51, Tenmile Freight the largest balance. Utilisation 91 percent, Crew Foxtrot fully committed.

Buyers are chasing acknowledgements on a fixed cadence now, and the cadence has already caught two silent date pushes. Supplier on-time delivery came in at 92 percent. Owen Castellanos issued the first supplier scorecards. Annual

planning was folded into this session. Next year's plan was already signed. Tessellate Controls missed two consecutive commit dates. On-time delivery was 92 percent, from 89 percent. Slipped deals 14, including Marrowfield Grocers. Toronto depot cleared its repair backlog. Tessellate Controls quoted lead times moved again. Priya Ramanathan and Dana Whitlock disagreed on sequencing.

Freight mode is a planning outcome rather than a logistics choice, and the air spend proves it. Expedite activity rose with the build plan of 72. Tessellate Controls missed two consecutive commit dates. Annual planning was folded into this session. Next year's plan was already signed. On-time delivery was 92 percent, from 89 percent. Coverage weeks on long-lead parts were extended. Uptime 97.3 percent, interventions every 344 hours. Toronto depot cleared its repair backlog. Bookings 204 units across 118 pursuits. Elk Grove carries 20 of the queued fleet.

Expedite spend is the honest measure of how well planning is working, and it went the wrong way this period. Supplier on-time delivery came in at 92 percent. Owen Castellanos issued the first supplier scorecards. Annual planning was folded into this session. Next year's plan was already signed. Coverage weeks on long-lead parts were extended. On-time delivery was 92 percent, from 89 percent. Scrap 25 thousand, Penrose Castings contributing. Build plan 72, actual 70, Controllers short. Yield 92 percent, scrap 25 thousand in Harness and electrical. Work in process 30 units, mostly Controllers.

One casting supplier has become a constraint on the pallet mover, and the tooling is theirs rather than ours. Owen Castellanos issued the first supplier scorecards. Supplier on-time delivery came in at 92 percent. Annual planning was folded into this session. Next year's plan was already signed. Penrose Castings improved after a site visit. Coverage weeks on long-lead parts were extended. Requisitions open 25, Nadia Oyelaran holding the list. Toronto depot cleared its repair backlog. Go-live cycle 54 days at Redlands. Sunil Achterberg owns the Controllers recovery.

Collections and days outstanding by account, Q4 2024

Account	Invoiced USD thousands	Received USD thousands	Days outstanding	Collection blocker
Bexley Pharmaceutical Wholesale	613	1982	54	purchase order number mismatch
Marrowfield Grocers	2090	1425	33	acceptance document unsigned
Pinebank Distribution	1655	1267	57	payables portal submission
Alder & Vance Retail	1393	1029	42	none
Northstar Cold Chain	1219	871	63	customer audit of milestones
Havering Logistics	958	792	48	credit note pending
Quillon Home Supply	784	554	69	purchase order number mismatch
Tenmile Freight	306	991	36	acceptance document unsigned
Bexley Pharmaceutical Wholesale - second site	1045	712	60	payables portal submission
Marrowfield Grocers - second site	827	633	45	none
Pinebank Distribution - second site	696	514	66	customer audit of milestones
Alder & Vance Retail - second site	609	435	39	credit note pending
Northstar Cold Chain - second site	479	396	54	purchase order number mismatch
Havering Logistics - second site	392	277	33	acceptance document unsigned
Quillon Home Supply - second site	204	660	57	payables portal submission

Material position

Work in process climbed because units are being held for a single missing part rather than because the line slowed down. Floor space in staging is now the binding limit. Kit shortages were concentrated in low-value parts. Annual planning was folded into this session. Next year's plan was already signed. Nadia Oyelaran owns the disposition decision. Work in process closed at 30 units. Headcount 183 of 194, Owen Castellanos reporting. Yield 92 percent, scrap 25 thousand in Harness and electrical. On-time delivery 92 percent, Brightwater Motion the outlier. Crew Foxtrot closed Tulsa without a second visit.

Excess and obsolete exposure sits almost entirely in the superseded sensor head, and the disposition decision has been deferred twice. Work in process closed at 30 units. Work in process moved from 28 to 30 units. Annual planning was folded into this session. Next year's plan was already signed. Nadia Oyelaran owns the disposition decision. Kit shortages were concentrated in low-value parts. Headcount 183 of 194, Owen Castellanos reporting. Burn 8910 thousand, runway 18 months. Days outstanding 51, Tenmile Freight the largest balance. Utilisation 91 percent, Crew Foxtrot fully committed.

Consignment terms with two suppliers would change the cash picture more than any other inventory action available. Work in process closed at 30 units. Buffer stock on the drive unit was raised deliberately. Annual planning was folded into this session. Next year's plan was already signed. Work in process moved from 28 to 30 units. Floor space in staging is now the binding limit. Slipped deals 14, including Marrowfield Grocers. Toronto depot cleared its repair backlog. Tessellate Controls quoted lead times moved again. Priya Ramanathan and Dana Whitlock disagreed on sequencing.

Inventory turns are a poor metric while the company is scaling, so the review looked at coverage weeks by commodity instead. Work in process moved from 28 to 30 units. Work in process closed at 30 units. Annual planning was folded into this session. Next year's plan was already signed. Kit shortages were concentrated in low-value parts. Floor space in staging is now the binding limit. Uptime 97.3 percent, interventions every 344 hours. Toronto depot cleared its repair backlog. Bookings 204 units across 118 pursuits. Elk Grove carries 20 of the queued fleet.

Receiving is holding returned parts that could be reworked, and nobody owns the decision to rework them. Work in process moved from 28 to 30 units. Work in process closed at 30 units. Annual planning was folded into this session. Next year's plan was already signed. Nadia Oyelaran owns the disposition decision. Kit shortages were concentrated in low-value parts. Scrap 25 thousand, Penrose Castings contributing. Build plan 72, actual 70, Controllers short. Yield 92 percent, scrap 25 thousand in Harness and electrical. Work in process 30 units, mostly Controllers.

Inventory is growing faster than build volume, and the growth is in long-lead components rather than in finished goods. Buffer stock on the drive unit was raised deliberately. Kit shortages were concentrated in low-value parts. Annual planning was folded into this session. Next year's plan was already signed. Work in process moved from 28 to 30 units. Work in process closed at 30 units. Requisitions open 25, Nadia Oyelaran holding the list. Toronto depot cleared its repair backlog. Go-live cycle 54 days at Redlands. Sunil Achterberg owns the Controllers recovery.

Runway scenarios tabled in the Q4 2024 review

Scenario	Horizon	Monthly burn USD thousands	Hiring posture	Runway months	Arming trigger
Upside	next quarter	3180	plan plus eight roles	16	two multi-site awards signed by quarter end
Upside	two quarters out	3320	plan plus eight roles	15	two multi-site awards signed by quarter end
Upside	four quarters out	3500	plan plus eight roles	14	two multi-site awards signed by quarter end
Middle, planning case	next quarter	2970	plan as approved	18	bookings within ten percent of plan
Middle, planning case	two quarters out	3110	plan as approved	17	bookings within ten percent of plan
Middle, planning case	four quarters out	3290	plan as approved	16	bookings within ten percent of plan
Downside	next quarter	2790	hiring pause on internal roles	21	two large awards leave the quarter
Downside	two quarters out	2930	hiring pause on internal roles	20	two large awards leave the quarter
Downside	four quarters out	3110	hiring pause on internal roles	19	two large awards leave the quarter
Severe	next quarter	2630	pause plus contract labour reduction	24	site queue shorter than nine sites
Severe	two quarters out	2770	pause plus contract labour reduction	23	site queue shorter than nine sites
Severe	four quarters out	2950	pause plus contract labour reduction	22	site queue shorter than nine sites

Uptime and interventions

Mean time between interventions is the number the service team manages to, and it moved in the right direction at every site but one. Tenmile Freight measures uptime on a stricter basis. Chandler South remains the weakest site on interventions. Annual planning was folded into this session. Next year's plan was already signed. Sunil Achterberg leads the joint remediation plan. Mean time between interventions was 344 hours. Headcount 183 of 194, Owen Castellanos reporting. Yield 92 percent, scrap 25 thousand in Harness and electrical. On-time delivery 92 percent, Brightwater Motion the outlier. Crew Foxtrot closed Tulsa without a second visit.

Facility conditions explain more of the intervention rate than any hardware factor, which took a year of data to establish convincingly. Interventions spaced 344 hours apart on the fleet median. Fleet uptime averaged 97.3 percent. Annual planning was folded into this session. Next year's plan was already signed. Mean time between interventions was 344 hours. Tenmile Freight measures uptime on a stricter basis. Headcount 183 of 194, Owen Castellanos reporting. Burn 8910 thousand, runway 18 months. Days outstanding 51, Tenmile Freight the largest balance. Utilisation 91 percent, Crew Foxtrot fully committed.

Service credits remain the sharpest feedback loop the company has, and the review looked at exposure rather than at incurred credits alone. Tenmile Freight measures uptime on a stricter basis. Sunil Achterberg leads the joint remediation plan. Annual planning was folded into this session. Next year's plan was already signed. Interventions spaced 344 hours apart on the fleet median. Chandler South remains the weakest site on interventions. Slipped deals 14, including Marrowfield Grocers. Toronto depot cleared its repair backlog. Tessellate Controls quoted lead times moved again. Priya Ramanathan and Dana Whitlock disagreed on sequencing.

Intervention data is only as good as what the operators record, and two sites have stopped recording anything useful. Chandler South remains the weakest site on interventions. Interventions spaced 344 hours apart on the fleet median. Annual planning was folded into this session. Next year's plan was already signed. Sunil Achterberg leads the joint remediation plan. Uptime reached 97.3 percent, from 97.0 percent. Uptime 97.3 percent, interventions every 344 hours. Toronto depot cleared its repair backlog. Bookings 204 units across 118 pursuits. Elk Grove carries 20 of the queued fleet.

A single navigation parameter set improved intervention rates at every site it was applied to, which is now the standard configuration. Chandler South remains the weakest site on interventions. Tenmile Freight measures uptime on a stricter basis. Annual planning was folded into this session. Next year's plan was already signed. Mean time between interventions was 344 hours. Uptime reached 97.3 percent, from 97.0 percent. Scrap 25 thousand, Penrose Castings contributing. Build plan 72, actual 70, Controllers short. Yield 92 percent, scrap 25 thousand in Harness and electrical. Work in process 30 units, mostly Controllers.

Intervention causes split roughly evenly between the robot, the facility and the operator, and only one of those three is ours to fix. Mean time between interventions was 344 hours. Chandler South remains the weakest site on interventions. Annual planning was folded into this session. Next year's plan was already signed. Uptime reached 97.3 percent, from 97.0 percent. Interventions spaced 344 hours apart on the fleet median. Requisitions open 25, Nadia Oyelaran holding the list. Toronto depot cleared its repair backlog. Go-live cycle 54 days at Redlands. Sunil Achterberg owns the Controllers recovery.

Points of disagreement

A minority view held that the second-source qualification was being rushed, and a further test was added because of it. Owen Castellanos and Nadia Oyelaran read the data differently. No consensus was reached; the Chief Executive Officer decided. Annual planning was folded into this session. Next year's plan was already signed. Dana Whitlock asked for the decision to be revisited. The disagreement is recorded rather than resolved. Headcount 183 of 194, Owen Castellanos reporting. Yield 92 percent, scrap 25 thousand in Harness and electrical. On-time delivery 92 percent, Brightwater Motion the outlier. Crew Foxtrot closed Tulsa without a second visit.

The plant manager objected to another plan revision inside the month and asked for a freeze window, which was granted in part. A further test was added because of the dissent. The minority view is preserved in the record. Annual planning was folded into this session. Next year's plan was already signed. The disagreement is recorded rather than resolved. Owen Castellanos and Nadia Oyelaran read the data differently. Headcount 183 of 194, Owen Castellanos

reporting. Burn 8910 thousand, runway 18 months. Days outstanding 51, Tenmile Freight the largest balance. Utilisation 91 percent, Crew Foxtrot fully committed.

Whether a deferral saves money depends on which function is counting, and both counts were tabled. No consensus was reached; the Chief Executive Officer decided. The disagreement is recorded rather than resolved. Annual planning was folded into this session. Next year's plan was already signed. The objection was noted against the decision. The minority view is preserved in the record. Slipped deals 14, including Marrowfield Grocers. Toronto depot cleared its repair backlog. Tessellate Controls quoted lead times moved again. Priya Ramanathan and Dana Whitlock disagreed on sequencing.

Disagreement was recorded rather than smoothed over, because the two positions rest on different readings of the same data. The minority view is preserved in the record. Dana Whitlock asked for the decision to be revisited. Annual planning was folded into this session. Next year's plan was already signed. A further test was added because of the dissent. The disagreement is recorded rather than resolved. Uptime 97.3 percent, interventions every 344 hours. Toronto depot cleared its repair backlog. Bookings 204 units across 118 pursuits. Elk Grove carries 20 of the queued fleet.

Finance and deployment disagreed about whether contract labour should be treated as a fixed or variable commitment. No consensus was reached; the Chief Executive Officer decided. The disagreement is recorded rather than resolved. Annual planning was folded into this session. Next year's plan was already signed. The minority view is preserved in the record. The objection was noted against the decision. Scrap 25 thousand, Penrose Castings contributing. Build plan 72, actual 70, Controllers short. Yield 92 percent, scrap 25 thousand in Harness and electrical. Work in process 30 units, mostly Controllers.

Two functions read the reliability data differently, and the review agreed to a single definition before arguing again. A further test was added because of the dissent. The objection was noted against the decision. Annual planning was folded into this session. Next year's plan was already signed. No consensus was reached; the Chief Executive Officer decided. The disagreement is recorded rather than resolved. Requisitions open 25, Nadia Oyelaran holding the list. Toronto depot cleared its repair backlog. Go-live cycle 54 days at Redlands. Sunil Achterberg owns the Controllers recovery.

Roadmap checkpoints against customer commitments, Q4 2024

Checkpoint	Product line	Committed date	Status	Dependency
Fleet orchestration release candidate	HS-200 tote handler	2025-02-26	on track	field data volume
Warehouse management adapter for the grocery host	HS-400 pallet mover	2025-03-20	at risk on dependency	test cell availability
Docking regression fix	Orchestrator fleet software	2025-04-11	on track	none
Controller revision B	Beacon WMS adapter	2025-05-03	at risk on dependency	none
Battery health telemetry	HS-200 tote handler	2025-05-25	on track	supplier firmware
Simulation coverage target	HS-400 pallet mover	2025-06-16	at risk on dependency	test cell availability
Operator training module	Orchestrator fleet software	2025-07-08	on track	certification of crew leads
Navigation parameter standardisation	Beacon WMS adapter	2025-07-30	at risk on dependency	field data volume
Fleet orchestration release candidate - second increment	HS-200 tote handler	2025-08-21	on track	certification of crew leads
Warehouse management adapter for the grocery host - second increment	HS-400 pallet mover	2025-09-12	at risk on dependency	none
Docking regression fix - second increment	Orchestrator fleet software	2025-10-04	on track	certification of crew leads
Controller revision B - second increment	Beacon WMS adapter	2025-10-26	at risk on dependency	customer test window
Battery health telemetry - second increment	HS-200 tote handler	2025-11-17	on track	test cell availability
Simulation coverage target - second increment	HS-400 pallet mover	2025-12-09	at risk on dependency	none
Operator training module - second increment	Orchestrator fleet software	2025-12-31	on track	supplier firmware
Navigation parameter standardisation - second increment	Beacon WMS adapter	2026-01-22	at risk on dependency	test cell availability

Decisions, owners and dates recorded in Q4 2024

Dates were negotiated with the owners present, which removed the usual round of silent renegotiation afterwards. Sunil Achterberg, Director of Field Service, took the largest open item. One item closed with measured evidence. Annual planning was folded into this session. Next year's plan was already signed. Priya Ramanathan inherited an item after a role change. Owners were named and dates set against 2025-02-27. Headcount 183 of 194, Owen Castellanos reporting. Yield 92 percent, scrap 25 thousand in Harness and electrical. On-time delivery 92 percent, Brightwater Motion the outlier. Crew Foxtrot closed Tulsa without a second visit.

Three items carried forward from the previous session, one with a new owner after the original owner changed roles. The oldest item dates from two sessions back. Three items carried forward into 2025-02-27. Annual planning was folded into this session. Next year's plan was already signed. One item closed with measured evidence. Two due dates were changed in the room. Headcount 183 of 194, Owen Castellanos reporting. Burn 8910 thousand, runway 18 months. Days outstanding 51, Tenmile Freight the largest balance. Utilisation 91 percent, Crew Foxtrot fully committed.

The action list was pruned deliberately, because a list nobody believes is worse than a shorter list that gets done. The oldest item dates from two sessions back. Sunil Achterberg, Director of Field Service, took the largest open item. Annual planning was folded into this session. Next year's plan was already signed. One item closed with measured evidence. Owners were named and dates set against 2025-02-27. Slipped deals 14, including Marrowfield Grocers. Toronto depot cleared its repair backlog. Tessellate Controls quoted lead times moved again. Priya Ramanathan and Dana Whitlock disagreed on sequencing.

Items closed this period were closed with named evidence: a report, a signed document or a measured result. Three items carried forward into 2025-02-27. One item closed with measured evidence. Annual planning was folded into this session. Next year's plan was already signed. Two due dates were changed in the room. Owners were named and dates set against 2025-02-27. Uptime 97.3 percent, interventions every 344 hours. Toronto depot cleared its repair backlog. Bookings 204 units across 118 pursuits. Elk Grove carries 20 of the queued fleet.

Open and closed actions carried through Q4 2024

Reference	Item	Owner	Original due	Current due	Disposition
OR-0214	Obtain written lead-time commitment from Kestrel on DU-4	Marcus Peil	2025-01-29	2025-02-19	Closed; signed supply agreement addendum, two-source clause
OR-0611	Root cause the sensor head bond-line yield loss	Emil Farkas	2025-02-09	2025-03-02	Closed; eight-panel study and revised cure profile record
OR-1204	Qualify Brightwater Motion as second drive-unit source	Hollis Bramwell	2025-03-16	2025-04-06	Open; none yet
OR-1006	Replace affected sensor heads at live sites on schedule	Sunil Achterberg	2025-02-14	2025-03-07	Open, date revised; partial
OR-1220	Stage software releases across sites rather than at once	Griffin Tasse	2025-01-26	2025-02-16	Open; outstanding
OR-1223	Refresh compensation bands for field engineering	Renata Voss	2025-02-01	2025-02-22	In progress; outstanding
OR-1226	Re-cut the stage definitions in the pipeline report	Marguerite Do	2025-02-07	2025-02-28	Closed; decision recorded
OR-1229	Publish the supplier scorecard with weighted criteria	Hollis Bramwell	2025-02-13	2025-03-06	Open; outstanding
OR-1232	Reconcile the two uptime definitions with the customer	Owen Castellanos	2025-02-19	2025-03-12	In progress; outstanding
OR-1235	Rebuild the site cost model separating retrofit from greenfield	Marcus Peil	2025-02-25	2025-03-18	Closed; report filed
OR-1238	Split the reliability backlog by build	Tobias Nkemdirim	2025-03-03	2025-03-24	Open; outstanding

Reference	Item	Owner	Original due	Current due	Disposition
	vintage				
OR-1241	Read the tooling ownership clauses on four part numbers	Hollis Bramwell	2025-03-09	2025-03-30	In progress; outstanding
OR-1244	Measure onboarding time to productivity for crews	Renata Voss	2025-03-15	2025-04-05	Closed; decision recorded
OR-1247	Cost the second-shift ramp against overtime	Marcus Peil	2025-03-21	2025-04-11	Open; outstanding
OR-1250	Draft deposit terms for multi-site commitments	Ingrid Solberg	2025-03-27	2025-04-17	In progress; outstanding
OR-1253	Bring deal-level notes for the top ten pursuits	Marguerite Do	2025-04-02	2025-04-23	Closed; memo circulated
OR-1256	Reissue the deployment playbook network section	Priya Ramanathan	2025-04-08	2025-04-29	Open; outstanding
OR-1259	Set a plan freeze window inside the build month	Emil Farkas	2025-04-14	2025-05-05	In progress; outstanding